

Current Solution

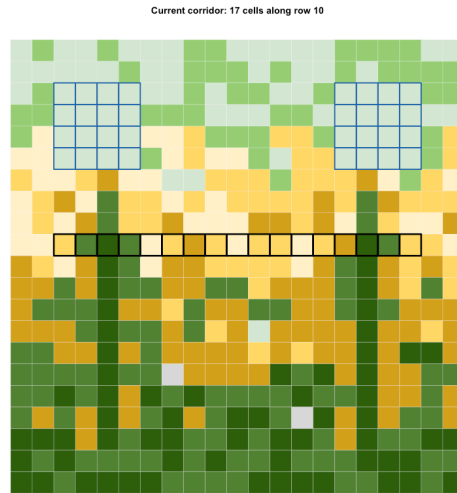


Figure 1: Current east-west corridor: 17 cells placed along row 10

The authority's current corridor (Figure 1) has two weaknesses. First, its design assumes fires travel northward from the southern scrubland, but the data shows realised winds blow from the west(70%) or south-east(30%), creating lateral pressure from directions the corridor does not block effectively. Second, the corridor commits the full 17-cell budget to a single uniform barrier rather than placing cells where they most reduce damage.

The corridor reduces expected damage by 71% relative to no firebreaks(13.56 to 4.01) but delivers only 0.56 damage reduction per cell against our proposed plan's 0.72, indicating inefficient use of budget.

Proposal

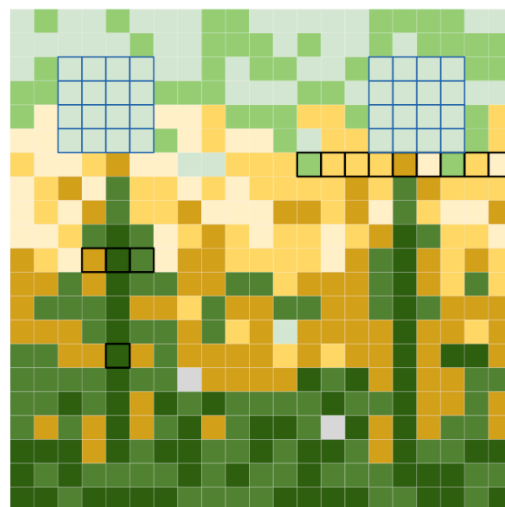


Figure 2: Recommended firebreak plan: 13 cells placed around the settlement and wetland

Figure 2 shows the recommended 13-cell firebreak plan (Model B, $B=13, \alpha=0.4$). It protects the settlement and wetland with 4 fewer cells than the current corridor while achieving comparable expected damage under realistic variation in wind and ignition, and delivers 29% more damage reduction per cell, making it the more resource-efficient choice.

The plan does not protect against fires ignited inside target cells, which remain the dominant source of damage in the worst 10% scenarios.

Scaffolding

STEP 1

MODEL

$V = \{(i, j) : \text{cell } (i, j) \text{ is vegetated}\} \subseteq \{1, \dots, 21\}^2$

$I = \{(21, j) \in V\} \rightarrow$ ignition cells: southernmost row 21

$T \subseteq V \rightarrow$ target cells: settlement or wetland

$N = V \cup \{s, t\} \rightarrow$ super source and super sink added to cells

$A = A_{\text{cells}} \cup A_s \cup A_t \cup \{(t, s)\} \rightarrow$ set of all directed edges

- $A_{\text{cells}} = \{(u, v) : u, v \in V, u \text{ is a neighbour cell of } v\}$
- $A_s = \{(s, u) : u \in I\}$
- $A_t = \{(v, t) : v \in T\}$

Parameters

Capacity of an edge (fire spread probability for the edge) :

Let $M = 10^6$ (proxy for infinity)

For each edge $e \in A$:

$$c_e = \begin{cases} p_0 \cdot (1 + p_{\text{veg},v}) \cdot (1 + p_{\text{den},v}) & e = (u, v) \in A_{\text{cells}}, v \notin T \\ M & e = (u, v) \in A_{\text{cells}}, v \in T \\ M & e \in A_s \cup A_t \cup \{(t, s)\} \end{cases}$$

where $p_0 = 0.25$, p_{veg} and p_{den} are the vegetation and density factors of the receiving cell v .

Decision Variables

$$f_e \geq 0 \quad \forall e \in A$$

Objective

$$\max f_{(t,s)}$$

Constraints

Capacity Limit & Non-negativity: Fire flow is limited by the fire spread capacity an edge can carry

$$0 \leq f_e \leq c_e \quad \forall e \in A$$

Fire Flow Conservation: All the fire that goes out from a cell is equal to the fire that comes into the cell

$$\sum_{(v,w) \in A} f_{(v,w)} - \sum_{(u,v) \in A} f_{(u,v)} = 0 \quad \forall v \in N$$

We model fire reachability as a max-flow problem on the grid network. Each adjacent cell pair produces two directed edges with different capacities, since the spread probability $p_{\text{burn}}(u,v)$ depends on the receiving cell v . Under Step 1's no-wind assumption, $p_{\text{wind}} = 1$ drops out. The max-flow problem is solved separately for each target. Shadow price on the capacity constraints identify saturated edges at the optimum i.e. the bottleneck edges forming the minimum-cut. We derive the firebreak proposal by clearing the receiving cells of these edges, since they remove all edges incident to them.

We use a circulation model with a virtual source s , sink t , and return arc (t,s) , maximising flow on the return arc. Edges entering target cells are given infinite capacity ($M = 10^6$) so they cannot bind at the optimum, otherwise target cells would receive positive shadow prices and be falsely flagged as firebreak locations.

CODE

Plain Text

```

1  set NODES;
2  set ARCS within {NODES, NODES};
3
4  param capacity {ARCS} >= 0;
5  param source symbolic in NODES;
6  param target symbolic in NODES;
7
8  var flow {(i,j) in ARCS} >= 0;
9
10 maximize TotalFlow:
11     flow[target, source];
12
13 subject to CapacityLimit {(i,j) in ARCS}: flow[i,j] <= capacity[i,j];
14 subject to Conservation {v in NODES}:
15     sum {(v,j) in ARCS} flow[v,j]
16     - sum {(i,v) in ARCS} flow[i,v] = 0;

```

R

```

1
2  set.seed(1)
3  source("fire-simulator.r")
4
5  landscape = as.matrix(read.table("landscape.csv", sep=","))
6  targets = read.csv("targets.csv")
7  settlement = targets[targets$weight==10,]
8  wetland = targets[targets$weight==5,]
9  edges = spread_probabilities(landscape, wind_speed=0, wind_dir=0)
10 node_name = function(r,c) paste0(r,"_",c)

```

```

11
12 #' Write an AMPL data file for the max-fire-flow problem
13 #' @param edges data frame with from_row, from_col, to_row, to_col, p_burn
14 #' @param landscape integer matrix (21x21)
15 #' @param target_cells data frame with row, col, weight
16 #' @param filename character string, path to output .dat file
17 #' @return NULL (writes file)
18 write_dat = function(edges, landscape, target_cells, filename){
19   nr = nrow(landscape); nc = ncol(landscape)
20   veg = which(landscape != 0L, arr.ind=TRUE)
21   all_nodes = c("s", "t", node_name(veg[,1], veg[,2]))
22   source_nodes = node_name(nr, which(landscape[nr,] != 0L))
23   target_nodes = node_name(target_cells$row, target_cells$col)
24
25   sink = file(filename, open="w")
26   cat("data;\n\n", file=sink)
27
28   cat("set NODES :=\n", file=sink)
29   for (n in all_nodes) cat(" ", n, "\n", file=sink)
30   cat(";\n\n", file=sink)
31
32   cat("param source := s;\nparam target := t;\n\n", file=sink)
33
34   cat("set ARCS :=\n", file=sink)
35   for (k in 1:nrow(edges)){
36     cat("(", node_name(edges$from_row[k], edges$from_col[k]), ",",
37       node_name(edges$to_row[k], edges$to_col[k]), ") \n",
file=sink)}
38   for (n in source_nodes) cat("(s,", n, ") \n", file=sink)
39   for (n in target_nodes) cat("( ", n, ",t) \n", file=sink)
40   cat("(t,s) \n;\n\n", file=sink)
41
42   target_pairs = paste0(target_cells$row, "_", target_cells$col)
43   edge_dest = paste0(edges$to_row, "_", edges$to_col)
44   is_target = edge_dest %in% target_pairs
45
46   cat("param capacity :=\n", file=sink)
47   for (k in 1:nrow(edges)){
48     cap = ifelse(is_target[k], 1e6, edges$p_burn[k])
49     cat("[", node_name(edges$from_row[k], edges$from_col[k]), ",",
50       node_name(edges$to_row[k], edges$to_col[k]), "]", cap, "\n",
file=sink)}
51   for (n in source_nodes) cat("[s,", n, "] 1e6\n", file=sink)
52   for (n in target_nodes) cat("[", n, ",t] 1e6\n", file=sink)
53   cat("[t,s] 1e6\n;\n", file=sink)
54   close(sink)}
55

```

```

56 write_dat(edges, landscape, settlement, "settlement.dat")
57 write_dat(edges, landscape, wetland, "wetland.dat")

```

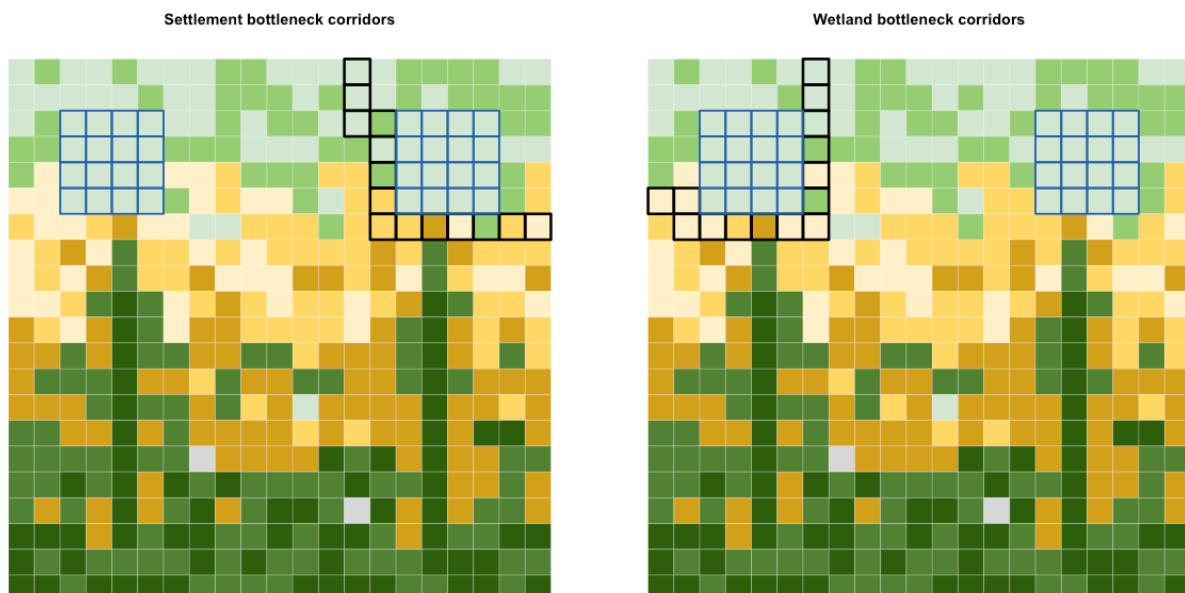
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```

1  option solver gurobi;
2  model step-1.mod;
3
4  data settlement.dat;
5  solve;
6  display TotalFlow;
7  #shadow prices are either 0 or 1 because of total unimodularity
8  #selecting dual=1 to identify edges in the minimum cut
9  display {(i,j) in ARCS: CapacityLimit[i,j].dual = 1} CapacityLimit[i,j].dual;
10
11 reset data;
12 data wetland.dat;
13 solve;
14 display TotalFlow;
15 display {(i,j) in ARCS: CapacityLimit[i,j].dual = 1} CapacityLimit[i,j].dual;

```

RESULTS



Bottleneck corridors by target

Maximum fire flow and number of bottleneck edges by target

	SETTLEMENT	WETLAND
MAXIMUM FIRE FLOW	7.605	6.615
COST	14 cells	14 cells

The maximum fire flow into the wetland (6.615) is lower than into the settlement (7.605), indicating that under the no-wind assumption the wetland is naturally less reachable from the southern ignition row. Both targets have 14 bottleneck edges whose receiving cells if cleared, would completely block the fire flow and protects the target.

For each target, the bottleneck edges form a single connected corridor wrapping the target on the southern and one lateral side, exactly the directions from which fire approaches under a southern ignition. The receiving cells of these edges constitute the proposed firebreak plan: a 14-cell corridor which would reduce max fire flow to 0.

DISCUSSION

1. The current corridor is a straight horizontal line and only intercepts fire travelling directly northward. The bottleneck corridors instead sit adjacent to each target, because fire originating in the south can curve around the corridor's open end and reach the wetland from its western side or the settlement from its eastern side. These lateral edges appear in the minimum cut with positive shadow prices, meaning fire pressure concentrates on them. The corridor does not intersect any of them.
2. Neither target is harder to protect than the other. Both require exactly 14 cells to fully block the fire. The settlement faces greater fire pressure, but the min-cut is achievable for both targets at the same price.
3. The model is deterministic as it treats spread probabilities as fixed edge capacities rather than as parameters of a stochastic process. Whether fire spreads along any given edge is actually a Bernoulli trial, so fire damage varies randomly even when conditions are identical. The max-flow value is an upper bound on reachability, not a prediction of what will burn. Model is also static as it captures whether fire can reach a target but not how quickly.

STEP 2

MODEL

We formulate the firebreak placement problem as a MILP. Both Model A and Model B share the same sets, cleared variable, weights, budget and objective structure. They differ only in how fire propagation is encoded.

V, I, T same as defined in Step 1

$A = \{(u, v) : u, v \in V, u \text{ is a neighbour cell of } v\}$

Model A: Binary Propagation

Under Model A, each edge is treated as either fully active or inactive based on a threshold $\tau \geq 0$. This is a worst-case model: every active edge carries fire with certainty.

Parameters

Damage weight of each cell:

$$w_{(i,j)} = \begin{cases} 10 & (i,j) \in T, (i,j) \text{ is settlement} \\ 5 & (i,j) \in T, (i,j) \text{ is wetland} \\ 0 & (i,j) \in V \setminus T \end{cases}$$

Active edge indicator:

$$\delta_a[u, v] = \mathbf{1}[p_{\text{burn}}(u \rightarrow v) > \tau] \in \{0, 1\} \quad \forall (u, v) \in A$$

Budget:

$$B \in \mathbb{Z}_{\geq 0}$$

Decision Variables

Cleared cell indicator:

$$\delta_c[i, j] \in \{0, 1\} \quad \forall (i, j) \in V$$

Burning cell indicator:

$$\delta_b[i, j] \in \{0, 1\} \quad \forall (i, j) \in V$$

Objective

Minimise the total weighted damage:

$$\min \sum_{(i,j) \in V} w_{(i,j)} \cdot \delta_b[i, j]$$

Constraints

$$\text{Budget: } \sum \delta_c \leq B$$

$$\text{Ignition: } \delta_b = 1 \quad \forall \text{ cell} \in I$$

$$\text{Target cell cannot be cleared: } \delta_c = 0 \quad \forall \text{ cell} \in T$$

$$\text{Cell can be either cleared or burned: } \delta_c + \delta_b \leq 1$$

Fire Propagation:

$$\delta_b[v] \geq \delta_a[u, v] \cdot \delta_b[u] - \delta_c[v] \quad \forall (u, v) \in A, v \notin I$$

When the incoming edge is inactive or the neighbour is unburnt, the RHS is at most 0, so $\text{burn}[v] = 0$ since we minimise damage. When v is cleared, the BurnOrCleared constraint forces $\text{burn}[v] = 0$ regardless of whether the edge is active.

Model B: Continuous Reachability

Parameters

$$w_{(i,j)}, B,$$

Compressed edge probability ($\alpha < 1$):

$$p[u, v] = p_{\text{burn}}(u \rightarrow v)^\alpha \in [0, 1] \quad \forall (u, v) \in A$$

Decision Variables

$$\delta_c, \text{ Fire reachability of each cell (continuous): } r[i, j] \in [0, 1] \quad \forall (i, j) \in V$$

Objective

Minimise the total weighted damage:

$$\min \sum_{(i,j) \in V} w_{(i,j)} \cdot r[i, j]$$

Constraints

$$\sum \delta_c \leq B$$

Ignition: $r = 1 \quad \forall \text{ cell} \in I$

$$\delta_c = 0 \quad \forall \text{ cell} \in T$$

Cell can be either cleared or burned: $r + \delta_c \leq 1$

Fire Propagation :

$$r[v] \geq p[u, v] \cdot r[u] - \delta_c[v] \quad \forall (u, v) \in A, v \notin I$$

The same logic applies with $r \in [0,1]$ continuous, so reachability decays multiplicatively along paths rather than cutting off sharply. Clearing still forces $r[v] = 0$ through the ClearedOrReachable constraint.

CODE

utils.r:

```

1  cell_name = function(r,c) paste0(r,"_",c)
2
3  write_cells = function(landscape,sink,set_name="CELLS"){
4      nr = nrow(landscape)
5      nc = ncol(landscape)
6      cat("set",set_name,":=\n",file=sink)
7      for (i in 1:nr){
8          for (j in 1:nc){
9              if (landscape[i,j] != 0L){
10                 cat(" ",cell_name(i,j),"\n",file=sink)}}}
11      cat(";\n\n",file=sink)}
12
13  write_edges = function(edges,sink,set_name="EDGES"){
14      cat("set",set_name,":=\n",file=sink)
15      for (k in 1:nrow(edges)){
16          from = cell_name(edges$from_row[k],edges$from_col[k])
17          to = cell_name(edges$to_row[k],edges$to_col[k])
18          cat("(" ,from," ",to,")\n",file=sink)}
19      cat(";\n\n",file=sink)}
20
21  write_ignition = function(landscape,sink){
22      nr = nrow(landscape)
23      nc = ncol(landscape)
24      cat("set IGNITION :=\n",file=sink)
25      for (j in 1:nc){
26          if (landscape[nr,j] != 0L){
27              cat(" ",cell_name(nr,j),"\n",file=sink)}}}
28      cat(";\n\n",file=sink)}
29
30  write_weight = function(landscape,targets,sink){
31      nr = nrow(landscape)

```

R


```

32     nc = ncol(landscape)
33     cat("param weight :=\n",file=sink)
34     for (i in 1:nr){
35         for (j in 1:nc){
36             if (landscape[i,j] != 0L){
37                 w = targets$weight[targets$row==i & targets$col==j]
38                 if (length(w)==0){ w = 0}
39                 cat(" [" ,cell_name(i,j), "]" ,w, "\n",file=sink)}}}
40     cat(";\n\n",file=sink)}
41
42 write_targets = function(targets,sink){
43     cat("set TARGETS :=\n",file=sink)
44     for (k in 1:nrow(targets)){
45         cat(" ",cell_name(targets$row[k],targets$col[k]), "\n",file=sink)}
46     cat(";\n\n",file=sink)}
47
48 write_edge_param = function(edges,values,param_name,sink){
49     cat("param",param_name,":=\n",file=sink)
50     for (k in 1:nrow(edges)){
51         cat("[",cell_name(edges$from_row[k],edges$from_col[k]), ",",
52             cell_name(edges$to_row[k],edges$to_col[k]),"]",
53             values[k], "\n",file=sink)}
54     cat(";\n\n",file=sink)}

```

Model A:

```

1  set CELLS;
2  set EDGES within {CELLS, CELLS};
3  set IGNITION within CELLS;
4  set TARGETS within CELLS;
5
6  param weight {CELLS} >= 0;
7  param B >= 0;
8  param active {EDGES} binary; #is edge active?
9
10 var cleared {CELLS} binary; #is cell cleared?
11 var burn {CELLS} binary; #is cell burning?
12
13 minimize TotalDamage:
14     sum {c in CELLS} weight[c] * burn[c];
15
16 subject to Budget:
17     sum {c in CELLS} cleared[c] <= B;

```

Plain Text

```

18
19 subject to Ignition {c in IGNITION}:
20     burn[c] = 1;
21
22 subject to NoTargetClearing {c in TARGETS}:
23     cleared[c] = 0;
24
25 subject to BurnOrCleared {c in CELLS}:
26     cleared[c] + burn[c] <= 1;
27
28 subject to Propagation {(m, c) in EDGES: c not in IGNITION}:
29     burn[c] >= active[m,c] * burn[m] - cleared[c];

```

Model B:

Plain Text

```

1  set CELLS;
2  set EDGES within {CELLS, CELLS};
3  set IGNITION within CELLS;
4  set TARGETS within CELLS;
5
6  param weight {CELLS} >= 0;
7  param B >= 0;
8  param compressed_p {EDGES} >= 0; # pburn^alpha for each edge
9
10 var cleared {CELLS} binary;      # is cell cleared? 1 or 0
11 var reachability {CELLS} >= 0, <= 1; # fire reachability of cell
12
13 minimize TotalDamage:
14     sum {c in CELLS} weight[c] * reachability[c];
15
16 subject to Budget:
17     sum {c in CELLS} cleared[c] <= B;
18
19 subject to Ignition {c in IGNITION}:
20     reachability[c] = 1;
21
22 subject to NoTargetClearing {c in TARGETS}:
23     cleared[c] = 0;
24
25 subject to ClearedOrReachable {c in CELLS}:
26     reachability[c] <= 1 - cleared[c];
27
28 subject to Propagation {(m, c) in EDGES: c not in IGNITION}:

```

```
29     reachability[c] >= compressed_p[m,c] * reachability[m] - cleared[c];
```

R

```
1  set.seed(1)
2  library(rAMPL)
3  source("utils.r")
4  source("fire-simulator.r")
5
6  landscape = as.matrix(read.table("landscape.csv",sep=",",))
7  targets = read.csv("targets.csv")
8  edges = spread_probabilities(landscape,wind_speed=0,wind_dir=0)
9
10 #' @param edges data frame with from_row,from_col,to_row,to_col,p_burn
11 #' @param landscape integer matrix (21x21)
12 #' @param targets data frame with row,col,weight
13 #' @param B integer budget
14 #' @param model character "A" or "B"
15 #' @param tau numeric threshold for active edges (Model A only)
16 #' @param alpha numeric compression parameter (Model B only)
17 #' @return list with damage,cleared,firebreaks matrix
18 solve_model = function(edges,landscape,targets,B,model,tau=0,alpha=0.4){
19
20     filename = paste0("step2_model",model,".dat")
21     sink = file(filename,open="w")
22     cat("data;\n\n",file=sink)
23
24     write_cells(landscape,sink)
25     write_edges(edges,sink)
26     write_ignition(landscape,sink)
27     write_weight(landscape,targets,sink)
28     write_targets(targets,sink)
29
30     if (model=="A"){write_edge_param(edges,as.integer(edges$p_burn >
31 tau),"active",sink)}
32     else{write_edge_param(edges,edges$p_burn^alpha,"compressed_p",sink)}
33
34     cat("param B :=",B,";\n",file=sink)
35     close(sink)
36
37     env = new(Environment,"/Applications/AMPL")
38     ampl = new(AMPL,env)
39     ampl$read(paste0("step2_model",model,".mod"))
40     ampl$readData(filename)
41     ampl$solve()
```

```

41   cleared = ampl$getVariable("cleared")$getValues()
42   damage = ampl$getObjective("TotalDamage")$value()
43   ampl$close()
44
45   list(damage=damage,cleared=cleared,firebreaks=get_firebreaks(list(cleared=cleared)))}
46
47   get_firebreaks = function(result){
48     df = as.data.frame(result$cleared)
49     cleared = df[df[,ncol(df)] > 0.5,]
50     if (nrow(cleared)==0) return(matrix(integer(0),nrow=0,ncol=2))
51
52     matrix(as.integer(do.call(rbind,strsplit(as.character(cleared[,1]),"_"))),ncol=2)}
53
54   evaluate_plan = function(landscape,firebreaks,target,N=1000){
55     if (!is.null(firebreaks) && nrow(firebreaks) > 0)
56       landscape = set_firebreaks(landscape,firebreaks)
57     mean(replicate(N,compute_damage(
58       simulate_fire(landscape,cbind(rep(21,21),1:21),0,0)$burned,target)))}
59
60   taus = c(0,0.1,0.2)
61   alphas = c(0,0.2,0.4,0.6,0.8)
62   budgets = c(0,5,13,17,19,20)
63   firebreaks_list = list()
64
65   resultsA_tau = do.call(rbind,lapply(taus,function(t){
66     res = solve_model(edges,landscape,target,B=0,model="A",tau=t)
67     data.frame(tau=t,damage=res$damage)}))
68
69   resultsA = do.call(rbind,lapply(budgets,function(b){
70     res = solve_model(edges,landscape,target,B=b,model="A",tau=0)
71     firebreaks_list[[paste0("A_B",b)]] <- res$firebreaks
72     data.frame(B=b,damage=res$damage,cleared=nrow(res$firebreaks)}))
73
74   resultsB_alpha = do.call(rbind,lapply(alphas,function(a){
75     res = solve_model(edges,landscape,target,B=13,model="B",alpha=a)
76     firebreaks_list[[paste0("B_a",a,"_B13")]] <- res$firebreaks
77     data.frame(alpha=a,damage=res$damage,cleared=nrow(res$firebreaks)}))
78
79   resultsB_budget = do.call(rbind,lapply(budgets,function(b){
80     res = solve_model(edges,landscape,target,B=b,model="B",alpha=0.4)
81     firebreaks_list[[paste0("B_B",b)]] <- res$firebreaks
82     data.frame(B=b,damage=res$damage,cleared=nrow(res$firebreaks)}))
83
84   corridor_landscape = set_firebreaks(landscape,cbind(rep(10,17),3:19))

```

```

83 corridor_edges = spread_probabilities(corridor_landscape,0,0)
84 corridor_A =
  solve_model(corridor_edges,corridor_landscape,targets,0,"A")$damage
85 corridor_B =
  solve_model(corridor_edges,corridor_landscape,targets,0,"B")$damage
86
87 evaluate_plan(landscape,NULL,targets)
88 evaluate_plan(landscape,firebreaks_list[["A_B13"]],targets)
89 evaluate_plan(landscape,firebreaks_list[["B_B13"]],targets)

```

RESULTS

Model A:

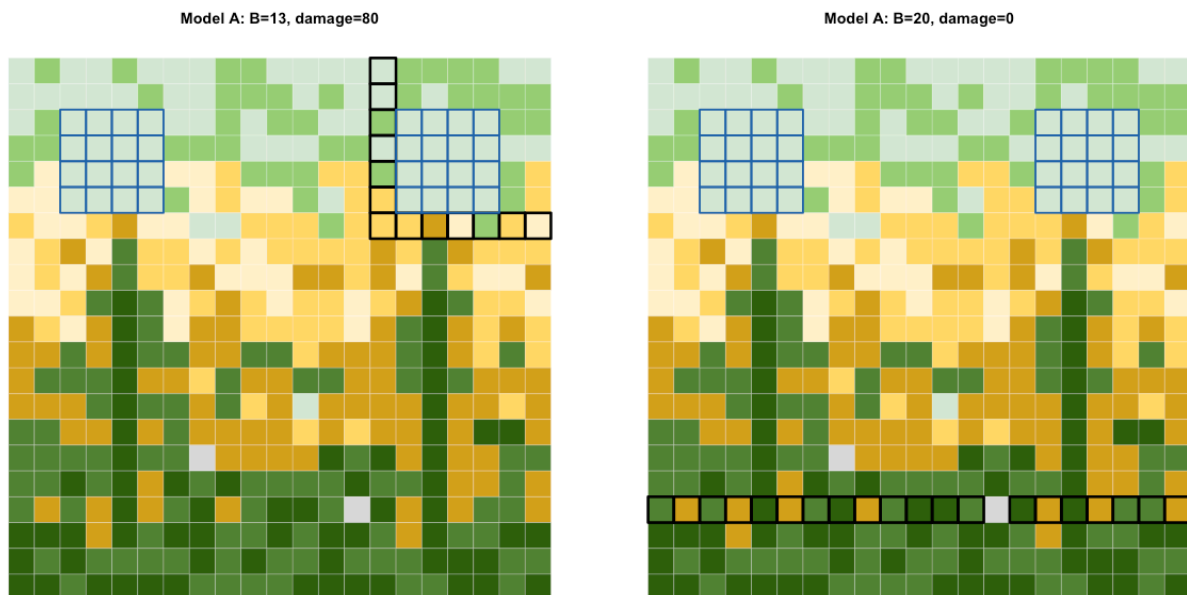


Figure 1 : Optimal firebreaks locations for Model-A for B = 13,20 at $\tau = 0, 0.1$

Table 1: Damage under Model A decreases in sharp step changes rather than smoothly:

Budget B	Damage
0-12	240
13-19	80
≥ 20	0

Model B:

Table 2: Different α values over current budget = 13

Alpha (α)	Damage
0	80
0.2	2.609
0.4	0.061
0.6	0.001
0.8	0.000

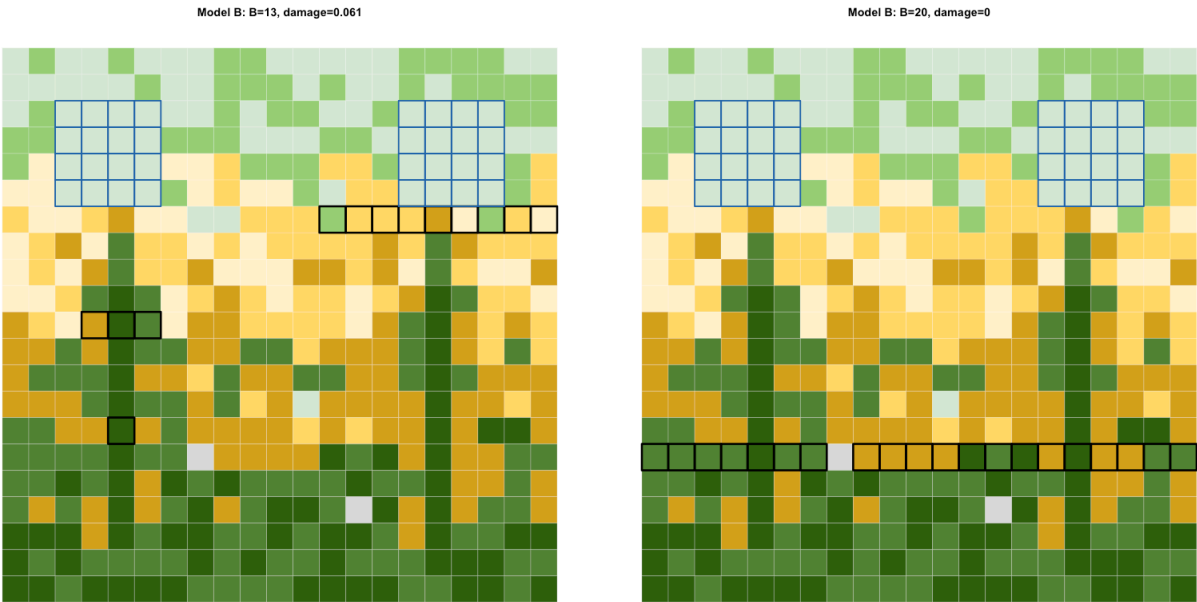


Figure 2: Optimal firebreak locations for Model-B for B=13,20 at $\alpha=0.4$

Table 3: Model B Damage over different budget values at $\alpha=0.4$

Budget B	Damage
0	0.34
5	0.21
13	0.06
20	0

Table 4: compares the current corridor against no firebreaks and the optimal B=13 plan under both models

Plan	cells	Model A damage	Model B damage
No firebreak	0	240	0.338
Current Corridor	17	240	0.127
B = 13	13	80	0.061

Table 5: Model predictions vs simulated damage under optimal firebreak plans

Plan	Simulated Damage	Model Prediction
No Firebreaks	16.43	240(A)/0.338(B)
Model A at B=13	4.21	80
Model A at B=20	0	0
Model B at B=13	3.04	0.061
Model B at B=20	0	0

DISCUSSION

Optimal firebreak locations and Effect of budget on damage

Model A:

For $\tau > 0.104$, enough edges deactivate to disconnect the targets, giving damage = 0 at any B. $\tau = 0$ reflects the worst case where every edge is active and carries fire with certainty; results at any τ between 0 and 0.104 coincide. Hence, we use $\tau = 0$ throughout as it is indifferent to choosing any other value in that range.

Under $\tau = 0$, every edge carries fire with certainty, so a firebreak only reduces damage if it completes a full cut between ignition and a target. Partial barriers are ineffective because fire routes around them.

Table 1's two drops correspond to the 2 minimum cuts: at B = 13 the solver places an L-shaped barrier (Figure-1) around the settlement's eastern and southern cells (wetland still burns, $16 \times 5 = 80$); at B = 20 it places a full horizontal barrier across row 16, disconnecting both targets (damage = 0). Between thresholds the surplus budget is unused, as no additional cut is achievable.

Model B:

Table 2 shows damage decreasing with α . We select $\alpha = 0.4$ because the compression is non-linear: different α values change the relative weights between edges, not just scale them uniformly. Lower alpha values inflate edges so aggressively that the model becomes overly pessimistic, while higher

values suppress reachability to near zero, underestimating true fire risk.

Figure 2 shows the $B = 13$ firebreak plan: a horizontal strip across row 7 on the right side, and a small cluster around rows 11-15 on the left. At $B = 20$, the firebreaks are placed on row 16, blocking every path from the south achieving 0 damage.

Shadow prices & Minimum Investment

The shadow price on the budget constraint is not formally defined for either model since both are MILPs and shadow prices require a continuous LP dual solution, which does not exist when variables are binary.

In **Model A**, the marginal value of one additional cell is 0 across most budget values, since the damage only drops at 2 critical thresholds. At $B=13$, one additional cell reduced damage by 160 and at $B=20$ by 80. These are threshold effects rather than stable shadow prices, a single cell only has value when it completes a full blocking corridor and is worth nothing otherwise.

In **Model B**, the shadow price can be approximated empirically. Table 3 shows that damage decreases as budget increases with diminishing returns:

- $B=0$ to 5: reduction = $0.34-0.21 = 0.13$, over 5 cells = **0.026 per cell**
- $B=5$ to 13: reduction = $0.21-0.06 = 0.15$, over 8 cells = **0.019 per cell**
- $B=13$ to 20: reduction = $0.06-0 = 0.06$, over 7 cells = **0.009 per cell**

This suggests that the first few firebreaks have the greatest impact, and each additional cell offers progressively less protection. Every cell contributes a small but non-zero reduction in expected damage, which contrasts with Model A where individual cells are either worthless or critical. Both models agree complete protection requires 20 cells. **Beyond that, additional spending helps neither.** For the authority, the practical takeaway is that 17 cells falls short of full protection under either model, so some residual risk is unavoidable.

Optimal Plan (Model A vs Model B)

At $B = 13$, the plan diverge. Model A protects only the higher-weight settlement (damage = 80) because with every edge active, any partial barrier is routed around and saves nothing, only fully disconnecting a target reduces damage and 13 cells can disconnect one target, not both. Model B instead distribute cells to partially reduce reachability to both (damage 0.061), as every cleared cell contributes under continuous propagation.

At $B = 20$, the plans converge on a horizontal barrier placement. Once there are enough cells to completely disconnect both targets, the binary/continuous distinction disappears.

Optimal Plan vs (Step 1 & Current East–West Corridor)

Current East–West Corridor (Table 4)

We evaluate the current east–west corridor by fixing its 17 cells as bare ground and computing the resulting damage under both models with zero budget. Table 4 displays the comparison.

Under **Model A**, the current corridor is equivalent to having no firebreaks at all, damage remains at 240 in both cases. The 2 open cells on each side create active bypasses that route fire around the corridor. $\tau = 0$ activates every edge, so any uncleared path from ignition to target lets fire reach it fully. $B=13$ performs better by saving the settlement.

Under **Model B**, the current corridor reduces expected damage by 62.4% relative to no firebreaks. However, the optimal for $B = 13$ plan achieves a 82% reduction using 4 fewer cells.

Both models therefore agree that the current corridor is an inefficient use of budget given conditions of these models as greater protection is offered at a much lower cost.

Step 1

Step 1's max-flow cannot express damage weights or a budget constraint, so it treats both targets as equally valuable and always solves for full protection (14 cells per target, solved separately). For the MILP at $B = 13$, Model A drops the lower-weight wetland, while Model B reduces reachability to both targets. Step 1 treats each target independently

The placement of firebreaks shows partial agreement between Step 1 and how Model A saves the settlement. Both concentrate barriers directly around the target, since both identify minimum cuts: Step 1 explicitly through max-flow and Model A implicitly since under $\tau = 0$ full protection requires a disconnecting set and the budget minimisation selects the smallest one. Model B places cells further south across the network because continuous reachability rewards any reduction in spread probability along a path, not only full disconnection.

Fire Simulator and Validation (Table 5)

Each damage value is the mean over 1000 replications. Table 5 reveals Model A over-predicts at $B=13$ since it represents a worst-case scenario. Model B under-predicts since multiplicative hops make reachability small relative to true stochastic spread. At $B=20$, both models predict simulated damage = 0 correctly, confirming complete protection is reliable at sufficient budget. Despite both model's poor prediction, Model B reduces damage 28% more than A, producing a better plan. This shows that prediction accuracy and plan quality are independent. The no-firebreaks baseline confirms both plans substantially outperform doing nothing.

Different Ignition Scenarios

The MILP is optimised for southern front ignition with no wind, so it only blocks fire approaching from the south. Under asymmetric wind, optimal firebreak locations shift; a single-cell ignition in the north would bypass the southern barriers entirely. The plan therefore performs well only in the scenario it was designed for.

Equal damage weights

Under equal weights, the plan $B=13$ would change, the solver would have no reason to favour one target and would seek a plan that balances protection across both. This reveals that the MILP does not treat targets neutrally, it implicitly prioritises whichever target generates the highest damage reduction per cleared cell.

STEP 3

MODEL

We estimate expected weighted damage under uncertain ignition, wind speed and direction by Monte Carlo: $E[D] \approx \frac{1}{N} \sum_{i=1}^N D_i$, where D_i is the Damage from sample i .

We use $N = 10000$, which gives Confidence Intervals tight enough that the ranking of the 3 plans is stable across different seeds.

We also report CVaR at $\alpha = 0.9$, estimated as the mean of damages in the worst 10% of samples $CVaR_{\alpha}(D) = E[D \mid D \geq VaR_{\alpha}(D)]$

By the Central Limit Theorem, the sample mean is approximately normally distributed, so a 95% Confidence Interval takes the form below with $\hat{\sigma}$ being the sample standard deviation of the damages. For the CVaR CI we apply the same formula to the worst 10% of damages only.

$$\bar{D} \pm 1.96 \cdot \frac{\hat{\sigma}}{\sqrt{N}}$$

Wind Speed: $V \sim \text{Weibull}(k=2, A=7)$

Ignition: $I \sim \text{Uniform}(\text{over all vegetated cells})$

Wind Direction: $\Theta \sim \text{two von Mises distributions}$

Acceptance-Rejection

Target density:

$$f_{\Theta}(\theta) = 0.7 \cdot \frac{e^{3 \cos(\theta - 1.5\pi)}}{2\pi I_0(3)} + 0.3 \cdot \frac{e^{2 \cos(\theta - 0.75\pi)}}{2\pi I_0(2)}$$

Proposal $Y \sim \text{Uniform}(0, 2\pi)$

$$f_Y(\theta) = \frac{1}{2\pi}, \quad \theta \in [0, 2\pi)$$

We use $\text{Uniform}(0, 2\pi)$ because it covers the full circular support of f_{Θ} and is trivial to sample. The mixture distribution has 2 directions (west, south-east) with moderate concentration ($K=2,3$), so the density is bounded implying an envelope exists.

We require: $M \geq \frac{\text{target density}}{\text{proposal density}} = \frac{f_{\Theta}(\theta)}{f_Y(\theta)} \quad \forall \theta$

so the tightest bound is $M = \max_{\theta} 2\pi f_{\Theta}(\theta)$

To find M , we evaluate $2\pi f_{\Theta}(\theta)$ at 10,000 equally spaced points in $[0, 2\pi)$ and take the maximum.

$M = 2.912769$

Acceptance rate = accepted proposals / total proposals = 0.342009

Theoretical acceptance rate = $1/M = 0.3433159$

Candidates from the proposal $\theta \sim Y$, are accepted when:

$$U \leq \frac{2\pi f_{\Theta}(\theta)}{M}, \quad U \sim \text{Uniform}(0, 1)$$

A better proposal would match the shape of f_{Θ} more closely. For eg, a distribution concentrated near the two peaks so that f_{Θ}/f_Y stays low across all θ , reducing M and raising the acceptance rate above 34.3%. However, any such proposal is itself harder to sample than a uniform, so the efficiency gain trades off against the implementation cost.

CODE

```

1  set.seed(1)
2  source("fire-simulator.r")
3  source("step-2.r")
4
5  wind_dir_density = function(theta){
6    0.7 * exp(3 * cos(theta - 1.5*pi)) / (2*pi * besselI(3,nu=0)) +

```

R

```

7      0.3 * exp(2 * cos(theta - 0.75*pi)) / (2*pi * besselI(2,nu=0))}
8
9      thetas = seq(0,2*pi,length.out=10000)
10     M = max(wind_dir_density(thetas) * (2*pi))
11
12     sample_wind_dir = function(n,wind_dir_density,M){
13       samples = numeric(n)
14       proposals = 0
15       for (i in 1:n){
16         repeat{
17           theta = runif(1,0,2*pi)
18           u = runif(1)
19           proposals = proposals + 1
20           if (u <= (2*pi * wind_dir_density(theta))/M){
21             samples[i] = theta
22             break}}
23       return (list(samples=samples,num_proposals=proposals))}
24
25     sample_wind_speed = function(n){
26       rweibull(n,shape=2,scale=7)}
27
28     sample_ignition = function(landscape){
29       veg_cells = which(landscape > 0,arr.ind=TRUE)
30       veg_cells[sample(nrow(veg_cells),1),]}
31
32     result = sample_wind_dir(10000,wind_dir_density,M)
33     Acceptance_rate = length(result$samples) / result$num_proposals
34
35     monte_carlo = function(landscape,firebreaks,target,N,wind_dir_density,M){
36       if (!is.null(firebreaks) && nrow(firebreaks) > 0){
37         landscape = set_firebreaks(landscape,firebreaks)}
38       damages = numeric(N)
39       burn_count = matrix(0,nrow(landscape),ncol(landscape))
40       for (i in 1:N){
41         wind_dir = sample_wind_dir(1,wind_dir_density,M)$samples
42         wind_speed = sample_wind_speed(1)
43         ignition = sample_ignition(landscape)
44         burned = simulate_fire(landscape,ignition,wind_speed,wind_dir)$burned
45         damages[i] = compute_damage(burned,target)
46         burn_count = burn_count + burned}
47       return (list(damages=damages,burn_freq=burn_count/N))}
48
49     N = 10000
50
51     solveB13 = solve_model(edges,landscape,target,B=13,model="B",alpha=0.4)
52     firebreaks_B13 = solveB13$firebreaks
53     corridor_firebreaks = cbind(rep(10,17),3:19)

```

```

54
55 B13 = monte_carlo(landscape,firebreaks_B13,targets,N,wind_dir_density,M)
56 corridor =
monte_carlo(landscape,corridor_firebreaks,targets,N,wind_dir_density,M)
57 none = monte_carlo(landscape,NULL,targets,N,wind_dir_density,M)
58
59 damages          = B13$damages
60 damages_corridor = corridor$damages
61 damages_none     = none$damages
62
63 tail_B13         = damages[damages >= quantile(damages,0.9)]
64 tail_corridor    = damages_corridor[damages_corridor >=
quantile(damages_corridor,0.9)]
65 tail_none        = damages_none[damages_none >= quantile(damages_none,0.9)]
66
67 cvar = function(damages,alpha=0.9){
68   threshold = quantile(damages,alpha)
69   mean(damages[damages >= threshold])}
70
71 ci = function(x){
72   m = mean(x)
73   k = sd(x) / sqrt(length(x))
74   c(lower=m - 1.96*k,estimate=m,upper=m + 1.96*k)}

```

RESULTS

We select the Model-B plan at B=13 from Step 2: it uses continuous probabilistic reachability, reflecting stochastic fire spread more faithfully than Step 1's min-cut (no budget awareness) or Model A's binary worst-case propagation. This makes Model-B the most appropriate plan to evaluate under uncertain wind and ignition.

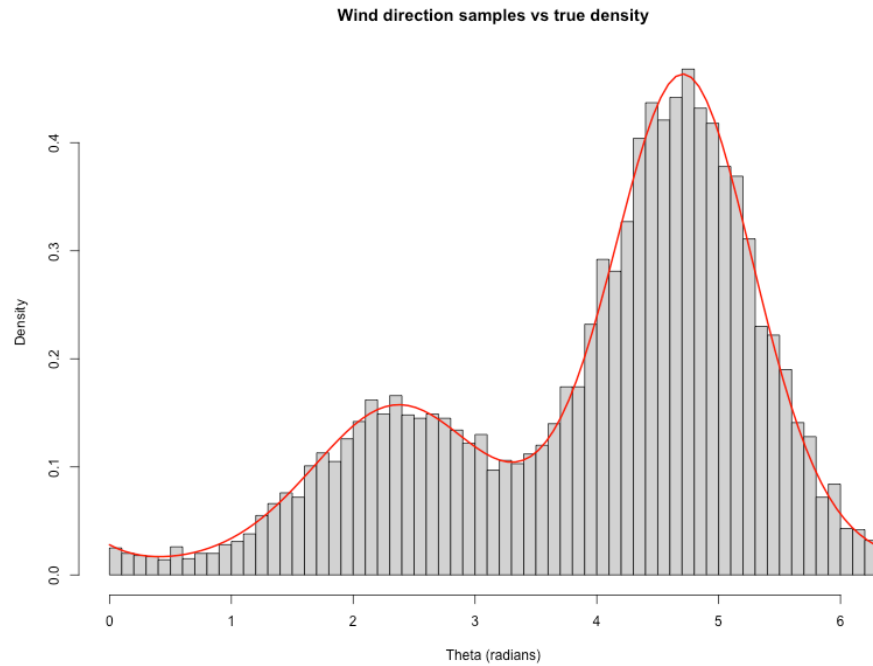


Figure 1: Verification of sampler producing correct density

Table 1: Expected damage, CVaR_{0.9} 95% CIs for the 3 plans across 10000 Monte Carlo samples.

Plan	Mean (Expected Damage)	SD	CVaR _{0.9}	Mean CI _{0.95}	CVaR _{0.9} CI _{0.95}
No firebreaks	13.56	21.16	60.31	13.14 - 13.97	59.12 - 61.51
Current Corridor	4.01	11.43	23.64	3.78 - 4.23	22.74 - 24.54
Model B, B=13	4.19	10.11	27.93	3.99 - 4.38	27.03 - 28.84

The corridor achieves the lowest expected damage and CVaR_{0.9}, though it uses 17 cells against Model B's 13. Model-B achieves mean damage close to the corridor but has a noticeably higher worst-case risk (27.93 vs 23.64), indicating weaker protection in the worst 10% of scenarios. Both plans drastically outperform no firebreaks on all measures.

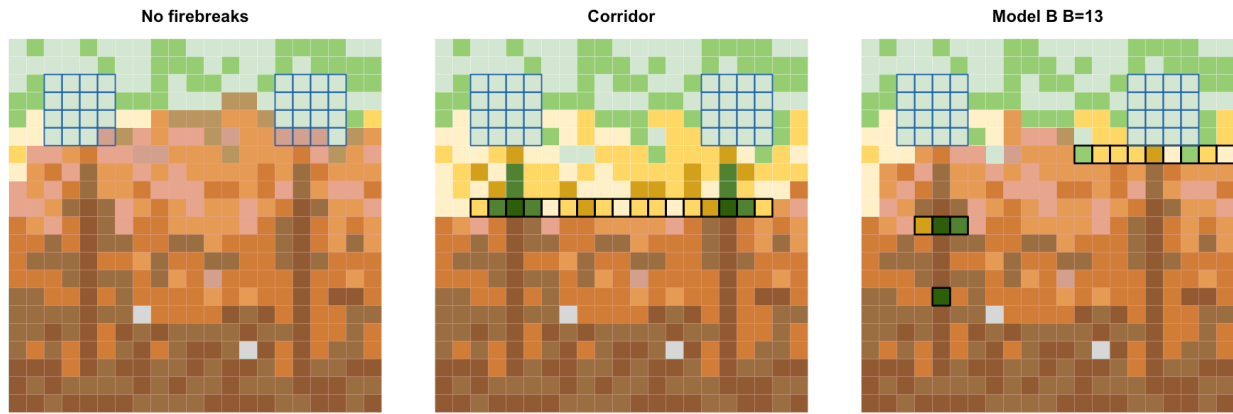


Figure 2: Model B protects the settlement but exposes the wetland; the corridor blocks fire north of row 10 almost entirely. Each cell in the plot is one that caught fire in more than 10% (>1000) of the 10,000 simulated scenarios.

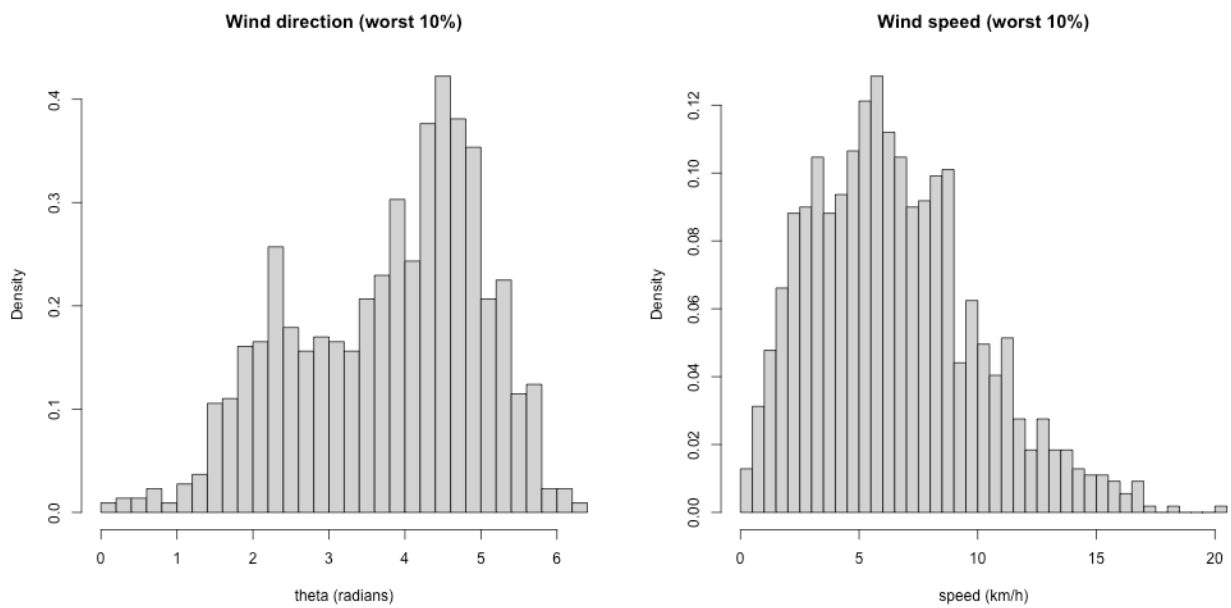


Figure 3:

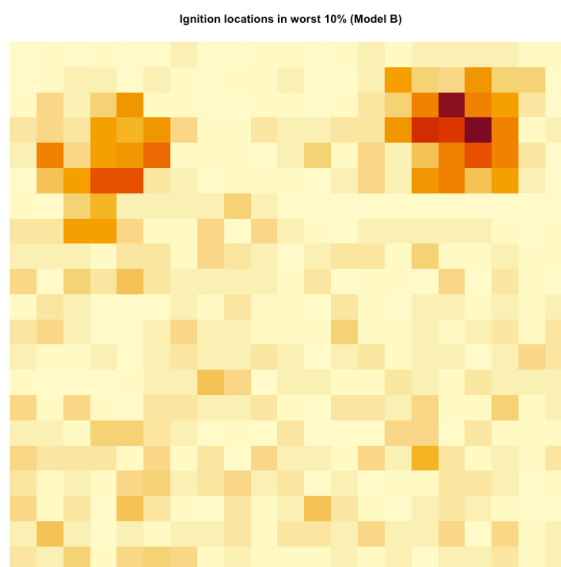


Figure 4: Ignition locations causing worst-case damage for Model-B cluster near the targets

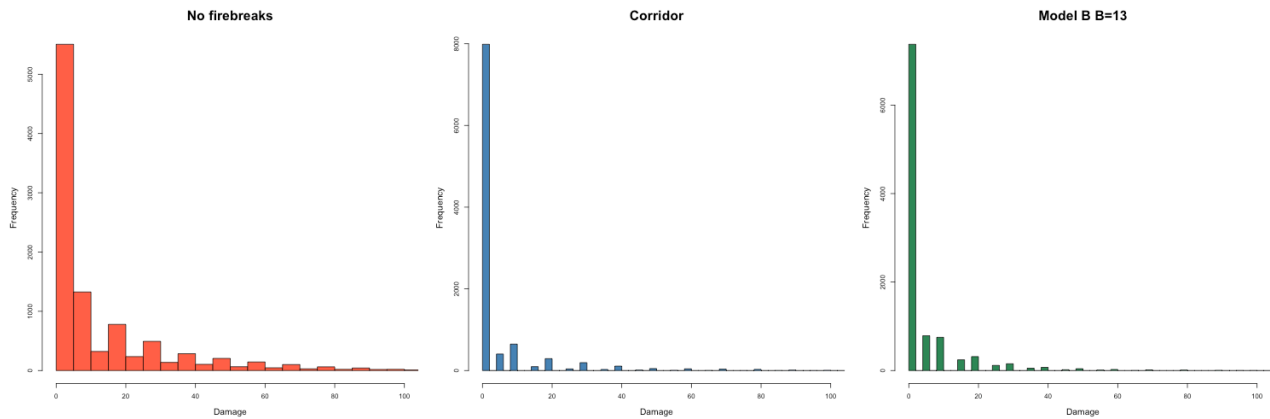


Figure 5: Positively-skewed damage distributions across scenarios

DISCUSSION

1. Worst-case conditions for Model B: The wind direction and wind speed distribution in the worst 10% (Figure 3) mirrors their respective density functions. Wind direction has a dominant peak at 4.5 radians ($1.5\pi = 4.71$), implying wind blowing from the west dominates the 10% worst samples. Another smaller peak is at 2.5 radians (0.75π). Wind speed also peaks around 7km/hr with no unusual bias in the worst case. Model B therefore has no specific wind-direction or wind-speed vulnerability.

The defining feature of the worst 10% is ignition location. Figure 4 heat-map shows that in the worst 10% cases, the ignition location was mostly inside the settlement itself. This reflects the deterministic optimisation of Model B, it was solved for a southern-front ignition with no wind, so its cell placement is optimised to defend against fires advancing from the south. When the fire ignites inside the settlement, Model B offers no protection. Since, the firebreaks defend the south boundary and damage accumulates immediately on high-weight target cells.

2. Targeted placement (Model B, 13 cells) achieves nearly the same expected damage as broad coverage (Corridor, 17 cells), 4.19 vs 4.01, despite using 4 fewer cells. Per-cell damage reduction relative to no firebreaks is 0.72 for Model B vs 0.56 for the corridor. The corridor's lower $CVaR_{0.9}$ (23.64 vs 27.93) comes from having more cells spread across more of the landscape, not from Model B's cells being poorly placed. As noted above, Model B's worst scenarios are driven by ignitions inside target cells, which no exterior firebreak plan can prevent. For the fire authority, this suggests that Model B's targeted placement delivers most of the corridor's protection at a lower cost, making targeted placement attractive when budget is constrained and deterministic assumptions are reasonable, while broad coverage is preferable when robustness to uncertainty is the priority.

3. If wind were known in advance, cells would be placed where fire actually approaches each target. Under westerly wind (70% of cases) the settlement faces higher risk from the west; under south-east wind (30%) the wetland faces higher risk from the south-east. Model B places a southern barrier, which matches neither case. For fires starting outside the targets, the wind-aware plan would perform noticeably better than Model B. However, the worst 10% of scenarios are driven by ignitions inside the targets, which wind knowledge cannot prevent, so overall improvement is capped.

4. Figure 5 shows that the damage distribution for all 3 plans has a dominant spike at 0, with the remaining mass spread in a heavy right tail. Most scenarios cause no damage because fire never reaches a target, the mean is therefore pulled down by these zeros even though the tail produces damages above 100. For Model B, expected damage is 4.19 but $CVaR_{0.9}$, nearly 7x higher. When the distribution is this skewed, the mean hides the outcomes the authority is most concerned about, while CVaR directly quantifies them. CVaR is therefore the more informative criterion for fire authority decisions, though mean remains useful as a summary of typical performance.

5. N must be large enough that the CIs are narrow enough to rank the plans. With $N=10000$, 'no firebreaks' has a CI of [13.14, 13.97] which does not overlap with either firebreak plan, so the main conclusion that 'firebreaks help' is fully settled. However, Model B and the Corridor have overlapping CIs in the range [3.99, 4.23], meaning more samples would be needed to confidently rank them on expected damage.

The choice of N depends on 3 factors: how spread out the damage values are (higher variance means wider CIs), whether we estimate the mean or CVaR (CVaR only uses the worst 10% of samples so needs 10x more N to match the Mean's precision) and how close the true damages of two plans are. For example, the gap between no firebreaks (13.56) and Model B (4.19) is 9.4, which is easily resolved at $N=10000$, but the gap between Model B (4.19) and Corridor (4.01) is only 0.18, which is smaller than the current CI half-widths of 0.2, so much larger N would be required to rank them confidently.

Discussion

Step 2's MILP improves on max-flow by incorporating a budget constraint and weighting targets by damage. This is also the source of Model B's exposure under uncertainty. Model B concentrates cells on settlement defence, which is rational given its weight, but this leaves the wetland structurally exposed. Figure 2 confirms this that under south-east wind, the wetland has almost protection.

Cell placement alone has a structural limit. Exterior firebreaks cannot prevent ignitions inside targets, which drive Model B's worst 10% of scenarios. The authority cannot reduce worst-case risk below $CVaR_{0.9}=27.93$ without non-firebreak interventions such as ignition prevention or early detection. Plan quality and prediction accuracy are distinct. Both models predict poorly against the simulator (Table 6), yet Model B's plan performs better.

The current corridor's rationale ("stops fires from the south") holds under the narrow assumptions for which it was designed, but fails the broader test. Our analysis shows the corridor's main advantage is 4 extra cells, not superior placement. Given this, we recommend Model B at $B = 13$, accepting its wetland risk as the explicit cost of cell-efficiency.

If seasonal shifts reversed the wind split toward south-easterly dominance, worst-case risk would concentrate on the wetland side where Model B is weakest, likely erasing its efficiency advantage. Historical wind data over multiple fire seasons would be needed to assess this risk properly.

AI Declaration

I formulated Step 1 and Step 2 Model-A independently using the Lecture 5 and Lecture 3 case studies, verifying Model-A constraints by testing all binary variable combinations. Model-B extended Model A. Proposal, current solution and discussion arguments are my own. I used Claude as a thinking partner to sanity-check my formulations and to revise A-R sampling from Lecture 4. I co-wrote some R code with Claude, mainly function structuring, roxygen comments, plots, LaTeX conversion of my hand-derived equations for Cadmus, and compressing code to meet the word limit.

